

PV Inverter Reliability Assessment: Methodology and Case Study

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Abstract—Reliable operation of photovoltaic (PV) inverters is critical for consistent power generation and minimizing maintenance costs in solar energy systems. An accurate reliability assessment plays a key role in optimizing system design, planning maintenance strategies, and enhancing operational efficiency. However, traditional assessment methods often rely on generic failure rates and simplified ambient conditions, limiting their accuracy by overlooking specific environmental stressors and mission profiles. To overcome these limitations, this paper introduces a comprehensive, application-oriented methodology that integrates tailored Physics-of-Failure (PoF) models with realistic, location-specific mission profiles. The proposed framework begins with a component-level evaluation of PV inverter reliability and is validated through a detailed case study using real-world field data. Results show a marked improvement in prediction accuracy over conventional approaches. Notably, the study underscores the importance of considering relative humidity—alongside temperature—as a critical factor in accelerating degradation, particularly in humid environments.

Index Terms—PV Inverter, Photovoltaic, Reliability Assessment

I. INTRODUCTION

Historical field data indicate that the reliability of power electronics components falls short of expectations [1]–[7]. In photovoltaic (PV) applications, PV inverters are often identified as the weakest link affecting system longevity, contributing between 43% and 70% of the O&M events in PV power plants [6], [7]. Therefore, the reliability assessment of PV inverters is critical for evaluating the reliability of the PV system. Reliability assessment is inherently a probabilistic problem focusing on evaluating the distribution of failures over time. Traditionally, metrics such as mean-time-to-failure (MTTF) and mean-time-between-failure (MTBF) have been extensively discussed in the literature and product documentation [8]. However, these metrics often assume a constant failure rate over the operational lifetime [9], which can lead to misleading conclusions in accurately predicting real-world reliability.

The modern mainstream approach to assessing the reliability of PV inverters involves Physics-of-Failure (PoF) models, which directly relate environmental and operational stressors to known failure mechanisms in critical components. Recent research related to inverter reliability assessment has emphasized the importance of integrating PoF models with real-world mission profiles. Nagarajan et al. [10] proposed an integrated PoF methodology, including multi-disciplinary

simulations and mission-profile translations, to enable accurate reliability assessment. Ma et al. [11] demonstrates that relying on typical average year mission profiles can lead to overestimations of inverter lifetime by about 32%, highlighting the critical need for location-specific mission data. [12] provides a comprehensive review of degradation and failure modes in PV modules, underscoring the importance of understanding component-level aging processes. [13] proposes a reliability evaluation framework that accounts for the interdependence of failure mechanisms, which can improve the accuracy of lifetime estimations. Lastly, [14] reviews various reliability assessment methodologies for grid-connected PV systems, highlighting practical approaches and challenges in real-world applications.

However, a methodological gap remains in effectively integrating tailored PoF models—selected based on relevant failure mechanisms—into practical assessments of component-level reliability in PV inverters using real-world mission profiles. To address the gap identified, this paper presents a comprehensive, application-oriented reliability assessment methodology that applies real-world mission profiles and failure mechanism-specific reliability modeling. The methodology is introduced in Section II, and a case study is performed in Section III using a real-world mission profile to demonstrate the validation of this methodology. Section IV summarizes the conclusion and future work.

II. METHODOLOGY

The methodology for assessing the reliability of PV inverters is introduced in this section. The approach involves four steps designed to systematically evaluate and quantify the PV inverter reliability. The flowchart of the proposed approach is summarized in Fig. 1.

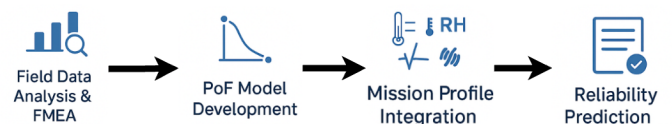


Fig. 1: Methodology Flowchart

- 1) In the first step, components exhibiting high failure rates are identified through the analysis of collected field data. Failure Modes and Effects Analysis (FMEA) is

employed to determine component failure mechanisms, calculate failure rates, and prioritize risks based on their potential impact. Critical components within the PV inverter are highlighted.

- 2) The second step involves developing and analyzing PoF reliability models for each critical component identified in the previous step. These reliability models are chosen according to the specific failure mechanisms observed, based on the literature review. Reliability model coefficients and parameters are determined through empirical testing, datasheet specifications, or comprehensive literature reviews. This ensures that the models accurately reflect component behavior under real operational conditions.
- 3) In the third step, accurate, multiphysics-based stress profiles—including electrical, thermal, and environmental stressors—for the high-failure-rate components identified in the first step have been generated. To achieve this, internal environmental conditions, specifically component surface temperature and relative humidity within the inverter enclosure, are first determined. These internal parameters provide a more precise basis for reliability assessment compared to ambient conditions, as they directly reflect the thermal and moisture stresses experienced by the components. Since PV inverters predominantly operate during daytime, only daytime data are considered. With these foundational internal stressors established, additional degradation factors such as operating voltage, current ripple, and mechanical vibration are integrated to build a comprehensive, dynamic mission profile. This profile naturally fluctuates over time, influenced by variations in environmental conditions—for instance, solar irradiance changing with the sun's position throughout the day, accompanied by corresponding temperature and humidity shifts. To ensure compatibility with the chosen reliability models, these dynamic stressors are ultimately converted into effective constant stress conditions, as described by Equation (1) [15].
- 4) The final step involves estimating the overall PV inverter lifetime by integrating the selected reliability models with the converted effective constant mission profiles. By evaluating each critical component under representative stress conditions, the analysis identifies the weakest component—the bottleneck component of the system. Consequently, the estimated lifetime of this bottleneck component is taken as the predicted overall lifetime of the PV inverter.

$$\xi_{\text{effective}} \simeq \frac{1}{\gamma} \ln \left\{ \frac{\Delta t}{t_b - t_a} \sum_{i=1}^m [f(\gamma\xi)]_i \right\} \quad (1)$$

III. CASE STUDIES

A case study is conducted to demonstrate the methodology outlined in Section II, using a commercial PV inverter.

The following subsections describe the practical implementation of each step in the proposed framework. For each critical component, lifetime models are developed, and the corresponding coefficients are identified. Real-world mission profiles are applied to estimate component lifetimes across four representative geographic locations: Miami, FL (hot and humid), Phoenix, AZ (hot and dry), Fort Peck, MT (cold and dry), and Riyadh, Saudi Arabia (extremely hot and dry). These locations are selected to represent severe environmental conditions and assess their impact on component reliability. The resulting lifetime estimates are compared to identify the primary reliability bottleneck under diverse operating environments.

A. Step 1: Identification of Critical Components using FMEA

Brown et al. [16] conducted FMEA to identify the high-failure-rate components of the PV inverter based on field data and historical failure records. Fig. 2 shows the risk priority number (PRN) of each component. The cooling fan, capacitors, power modules, and PCB interface were identified with high RPN values and will be evaluated in the following steps. The failure of MOV is primarily due to the dynamic effects in the PV system, such as lightning and voltage surge, which is relatively standalone and has been well discussed in [17], and will be de-emphasized here.”

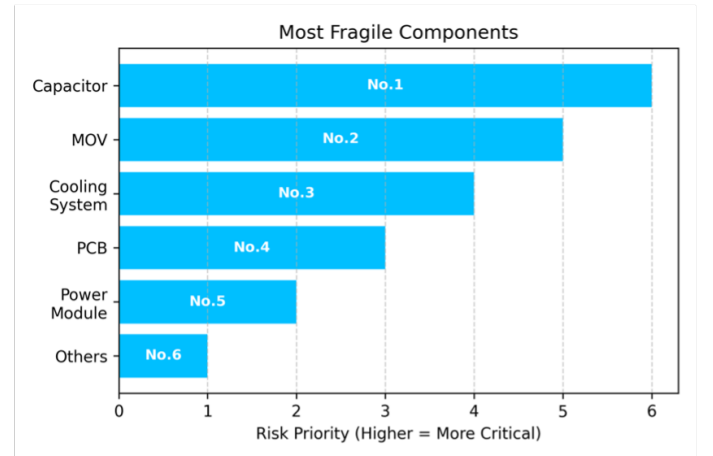


Fig. 2: Risk Priority Number of Each Component

B. Step 2: Reliability Models for Individual Components

Detailed reliability analyses were performed separately for each identified critical component, following the established procedure: starting with identifying the crucial failure mechanisms, and then deriving the reliability models and corresponding coefficients.

1) *Cooling Fan*: The fishbone diagram for cooling fan is shown in Fig. 3. As can be seen in the diagram, three main failure mechanisms lead to cooling fan failure: wear out, lubrication deterioration, and electrical and environmental stress. Of these failure mechanisms, wear out and lubrication deterioration affect the mechanical system, whereas electrical and

environmental stress affect the electrical system. Lubrication deterioration is the failure mechanism most observed in the field for the mechanical system [15]. These locations represent a range of severe environmental conditions used to assess their impact on component reliability. The resulting lifetime estimates are compared to identify the primary reliability bottlenecks across different operating environments. Furthermore, electrical and environmental factors are the primary stressors in the electrical system that significantly influences the lifetime of cooling fans. This failure mechanism directly affects the life of the fan controller and internal electronics (a small PCB integrated with the cooling fan).

Based on this analysis, it is determined that two lifetime models are necessary for cooling fans, one for the bearings of the mechanical system and one for the PCB and electronics of the electrical system (these components are highlighted in red in the diagram).

Targeting these mechanisms, the corresponding reliability models were derived from literature reviews [15], as shown in Equations (2) and (3). The coefficients are summarized in Table I.

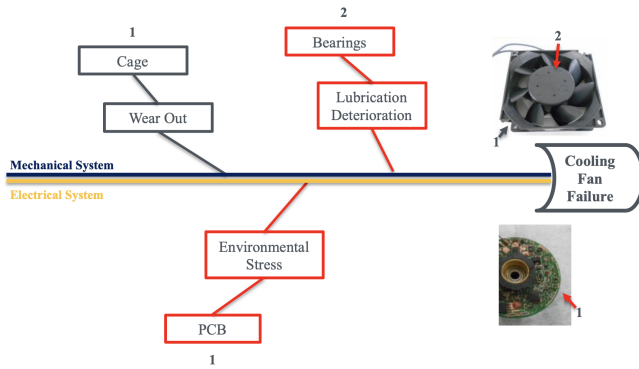


Fig. 3: Cooling Fan Fishbone Diagram

$$MTTF = A \cdot RH_{op}^n \cdot e^{\left(\frac{E_a}{K_B \cdot T_{op}}\right)} \quad (2)$$

$$L_{10} = A \cdot \exp(c \cdot T_{op}) \quad (3)$$

2) *Capacitor*: The aluminum (Al)-electrolytic capacitors are widely used as the DC-link and AC filter in the selected inverter. The reliability fishbone diagram is shown in Fig. 4. The failure modes, failure mechanisms and corresponding critical stressors are described below.

$$MTTF = L_0 \cdot \frac{V^{-\beta_1}}{V_0} \cdot 2^{\frac{T_0 - T}{10}} \quad (4)$$

For Al-Electrolytic Capacitors, electrolyte vaporization is the major wear out mechanism of small size Al-caps (e.g. snap-in type) due to their relatively high equivalent series resistance (ESR) and limited heat dissipation surface. For large size Al-caps, the wear out lifetime is dominantly determined by the increase of leakage current, which is associated with the

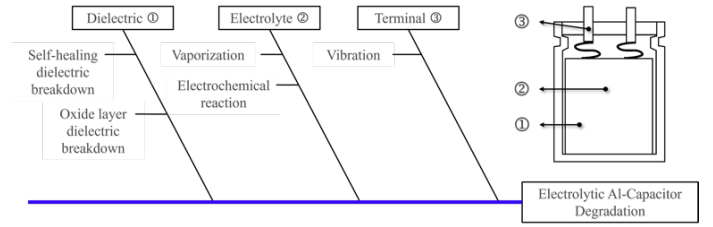


Fig. 4: Aluminum Electrolytic Capacitor Fishbone Diagram

electrochemical reaction of the oxide layer [18]. The reliability model was from a literature review [19], which is shown in (4). The range of coefficients are summarized in Table I.

3) *Power Module*: The fishbone diagram for power module degradation is shown in Fig. 5. In PV inverters, power module degradation is primarily caused by failures within the packaging. Key components such as bonding wires, solder joints, substrates, and metallization layers are particularly prone to wear and failure.

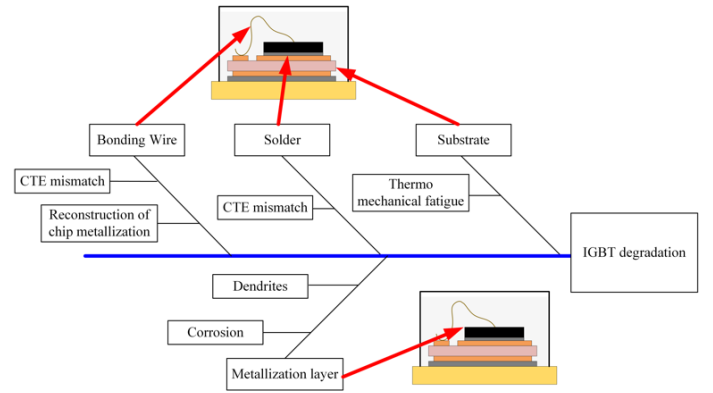


Fig. 5: IGBT Fishbone Diagram

The leading cause of IGBT packaging failure is thermo-mechanical fatigue, which occurs due to repeated temperature fluctuations. These fluctuations create a Coefficient of Thermal Expansion (CTE) mismatch, leading to the degradation of bonding wires, solder joints, or substrates. The primary stressors responsible for this degradation are maximum junction temperature and junction temperature swing [20]. To estimate the lifespan of IGBTs under these conditions, a modified Coffin-Manson equation is commonly used.

$$N_f = A \cdot \Delta T^{n_1} \cdot \exp\left(\frac{E_a}{K_B} \cdot \frac{1}{T_{max}}\right) \quad (5)$$

Another significant failure mechanism is temperature and humidity bias, which accelerates corrosion and dendrite formation inside the package. This degradation occurs due to prolonged exposure to heat and humidity. Peck's relationship is typically applied to evaluate the impact of these environmental factors on power module lifespan. Additionally, voltage and the resultant electric field would further accelerate the process, which should be considered. Therefore, in this case, key stressors considered for lifetime estimation include humidity, temperature, and voltage bias.

$$MTTF = MTTF_{\text{stress}} \cdot \left(\frac{RH_{\text{stress}}}{RH_{\text{op}}} \right)^{-n} \cdot \exp \left(\frac{E_a}{K_B} \left(\frac{1}{T_{\text{op}}} - \frac{1}{T_{\text{stress}}} \right) \right) \cdot \left(\frac{V_{\text{stress}}}{V_{\text{op}}} \right)^{-n_2} \quad (6)$$

The reliability model considering CTE mismatch and reconstruction of chip metallization is shown in (5), and the model considering corrosion and dendrites is shown in (6). The range of coefficients are summarized in Table I.

4) *PCB*: The fishbone diagram is shown in Fig. 6. PCB degradation is primarily caused by the failure of surface mount components that lack leads and are directly soldered onto the board. The components with large footprint are most vulnerable. Upon close examination of the target commercial inverter, the most affected components are decoupling capacitors and Metal Electrode Leadless Face (MELF) resistors. The main failure mechanisms in solder joints are thermomechanical fatigue and creep. Energy-density-based models are typically used to estimate PCB lifetimes under these conditions [21].

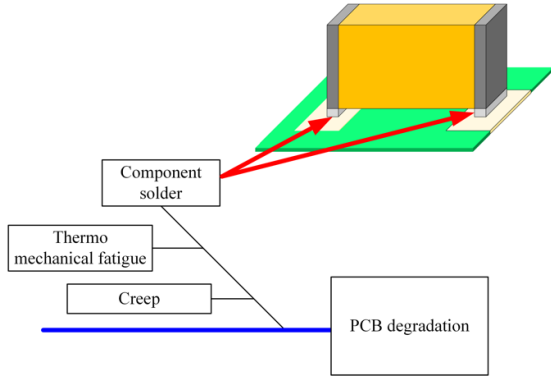


Fig. 6: PCB Fishbone Diagram

The corresponding reliability model was established in (7). Coefficients are summarized in TABLE I

$$N_{f,50\%} = \left(\frac{\Delta W}{\epsilon_f} \right)^{c_{fatigue}} \quad (7)$$

C. Step 3: Incorporation of Mission Profile

To support high-fidelity reliability estimation, the third step begins by determining the internal environmental conditions—specifically, the temperature at the component surface and the relative humidity within the inverter enclosure. These parameters form the foundation for calculating the stressors experienced by individual components. A data-driven model based on a random forest algorithm [26] is employed to predict internal temperature and humidity using measured ambient conditions as inputs. This approach achieves over 90% accuracy for temperature and 85% for relative humidity. Only daytime data are considered, as PV inverters operate predominantly during daylight hours. The predicted waveforms for internal temperature and humidity are shown in Fig. 7.

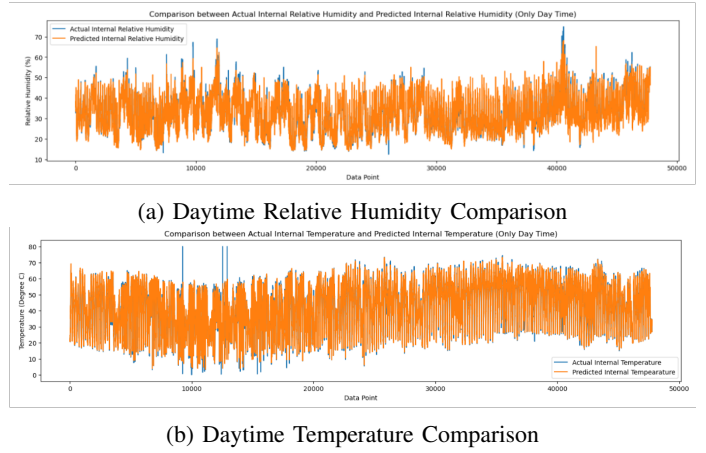


Fig. 7: Comparison of day time relative humidity and temperature

Building upon these internal environmental conditions, specific stressors for each critical component are extracted. For the cooling fan, both electrical and mechanical subsystems are analyzed. The electrical subsystem is primarily influenced by internal temperature and relative humidity, while the mechanical subsystem is dominated by steady-state operating temperature. For the aluminum electrolytic capacitor, the primary stressor is the operating voltage, which accelerates electrolyte degradation. The power module experiences degradation due to two dominant mechanisms: (1) corrosion and dendrite formation driven by temperature, humidity, and voltage bias; and (2) thermo-mechanical fatigue caused by repetitive temperature swings (ΔT) at the junction level. Lastly, the PCB interface is subjected to thermal cycling stress, with the temperature swing (ΔT) being the primary driver of fatigue in surface-mounted components and solder joints. The temperature swings are derived from the four-point rainflow cycle counting algorithms [27].

To ensure compatibility with the PoF-based reliability models, the time-series mission profiles derived from environmental and electrical conditions are converted into effective constant stressors using the transformation defined in Equation (1). Table II summarizes the resulting effective constant temperature and humidity stressors for each critical component across the four representative geographic locations. To simplify subsequent computations, only the stressors contributing to the worst-case scenario are retained, as some model coefficients fall within a range. Additionally, because the derived coefficients for the PCB reliability model are not directly correlated with temperature swings, no meaningful equivalent stressor can be determined; therefore, the corresponding effective constant temperature swings are not included in the table. These values serve as the input conditions for Step 4 to estimate component-level lifetimes.

D. Step 4: Lifetime Estimation

Using the effective constant stress levels in Table II, along with the reliability models in (2-7), the lifetime of individual

TABLE I: Summary of Acceleration Coefficients for Key PV Inverter Components

Component	Constant Coefficients	RH Coefficients	Temperature Coefficients	Operating Voltage Coefficients	Temperature Swing Coefficients	Source
Cooling Fan (Electrical)	0.001	-3.06	0.86	N/A	N/A	Acceleration Test
Cooling Fan (Mechanical)	671.027	N/A	-0.04	N/A	N/A	Manufacturer Datasheet
Aluminum Electrolytic Capacitor	N/A	N/A	N/A	[3, 5]	N/A	Literature review [19]
Power Module (CTE mismatch)	30,000	N/A	[0.2, 0.12]	N/A	[-3.22, 3.746]	Literature Review [22]
Power Module (Corrosion and dendrites)	0.7911	[2.66, 4.64]	[0.79, 0.9]	2.2	N/A	Acceleration Test & Literature Review [23], [24]
PCB	$1.2473 \cdot 10^{-5}$	N/A	N/A	N/A	-1.3	Literature review [25]

TABLE II: Effective Constant Stressors for Four Representative Locations

Location	Cooling Fan (Electrical)		Cooling Fan (Mechanical) Temp. (°C)	Capacitor (Aluminum) Temp. (°C)	Power Module		Power Module ΔT (°C)
	Temp. (°C)	RH (%)			Temp. (°C)	RH (%)	
Phoenix, AZ	49.20	18.56	35.31	92.05	43.93	23.63	25.21
Miami, FL	45.97	39.79	37.46	81.47	43.16	41.12	14.05
Fort Peck, MT	44.81	26.56	33.42	81.50	42.54	30.79	43.41
Riyadh, Saudi Arabia	54.13	17.32	44.86	91.75	50.56	20.83	24.73

components can be calculated, as shown in Table III.

E. Result Analysis

The estimated component lifetimes across the four representative locations reveal several key insights regarding PV inverter reliability:

- **Relative humidity (RH) is a critical stressor:** RH significantly accelerates component degradation, particularly in power modules and PCB interfaces. For example, components in humid environments such as Miami exhibit much shorter lifetimes due to moisture-induced degradation mechanisms, including corrosion and dendrite formation.
- **Modeling internal temperature and RH is essential:** It is noted that precisely estimating the internal environmental conditions—such as temperature and relative humidity—within the inverter enclosure is critical for capturing the actual stressors experienced by components. This internal perspective is expected to improve the accuracy of lifetime predictions and reliability assessments.
- **Geographic variability influences dominant failure mechanisms and design priorities:** The identified bottleneck components vary by location, emphasizing the importance of location-specific mission profiles and targeted reliability strategies. For example:
 - In **hot and dry climates** (e.g., Phoenix, Riyadh), *capacitors* are most vulnerable due to elevated temperature and voltage stress, suggesting a need for enhanced capacitor robustness.
 - In **humid environments** (e.g., Miami), *power modules* are more susceptible to corrosion and dendrite formation, indicating the importance of improved moisture protection.
 - In **cold and dry regions** (e.g., Fort Peck), *PCB interfaces* become the limiting factor due to thermal cycling-induced fatigue, calling for improved mechanical resilience.

IV. CONCLUSION

This paper presents a comprehensive and application-oriented methodology for assessing the reliability of PV inverters. The proposed approach integrates Physics-of-Failure (PoF) models—tailored to specific failure mechanisms and

integrated with location-specific mission profiles. The methodology involves: (1) identifying high-risk components through FMEA, (2) constructing component-specific reliability models, (3) reconstructing internal stressors based on environmental and operational conditions, and (4) estimating component lifetimes using effective constant stress levels derived from dynamic mission profiles.

A case study using a commercial PV inverter validates the effectiveness of this methodology. The study incorporates measured mission profiles from four representative geographic locations. The lifetime prediction results reveal several key findings:

- 1) **Relative humidity is a critical environmental stressor** and must be considered alongside temperature in reliability assessments.
- 2) **Accurately estimating internal temperature and RH** within the inverter enclosure is essential for capturing the true stress conditions experienced by components.
- 3) **Geographic location significantly influences the dominant failure mechanism**, and therefore, the weakest component varies with climate. This highlights the need for location-specific mission profiles and tailored design or maintenance strategies.

The methodology provides actionable insights that can inform inverter design improvements, environmental protection measures, and the development of more realistic accelerated life tests.

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TABLE III: Lifetime Summary for Four Representative Locations

Location	Cooling Fan (yrs)	Capacitor (yrs)	Power Module (yrs)	PCB Interface (yrs)	Earliest Failed Component
Phoenix, AZ	29.13	4.36	9.45	29.83	Capacitor due to voltage and temperature
Miami, FL	9.41	9.08	2.58	149.07	Power Module due to thermal cycling
Fort Peck, MT	29.13	9.07	5.90	8.47	PCB Interface due to humidity cycling
Riyadh, Saudi Arabia	27.09	4.45	8.13	36.83	Capacitor due to high temperature

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